

Accelerated Radiative Diffusion Problems

Original article: Synthetic Acceleration for Radiative Diffusion Calculations [1]

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1. Problem

The equations for radiative diffusion are

$$\frac{1}{c} \frac{\partial I}{\partial t} - \nabla \cdot D(\nu) \nabla I(\nu) + \rho \kappa(E) I(E) = \rho \kappa(\nu) \beta(\nu, T) \quad (1a)$$

$$\rho C_v \frac{\partial T}{\partial t} = \nabla \cdot K \nabla T - \nabla \mathbb{P} u + \rho \int_0^\infty \kappa(E') [I(E') - \beta(E', T)] dE' + Q, \quad (1b)$$

where Equation 1a is the radiative diffusion equation, where c is the speed of light, I is scalar intensity $I_\nu(x) = \int_{4\pi} \mathcal{I}_\nu(x, \Omega) d\Omega = cE$, D is the radiative diffusion constant $(3\rho\kappa)^{-1}$, κ is specific opacity [cm^2/g], and β is the angle-integrated Planck function, $\beta_\nu(T) = \int_{4\pi} B_\nu(T) d\Omega$.

Equation 1b is the heat diffusion equation with radiation and hydrodynamic effects, where ρ is material density, C_v is material heat capacity, K is thermal conductivity, $\mathbb{P}$ is the fluid pressure tensor, u is fluid velocity, and Q is heat generation per unit mass.

The energy domain can be discretized using N_g energy groups $G_i = [\nu_i, \nu_{i+1}]$ such that $\nu_0 = 0$ and $\nu_{N_g+1} = \infty$. Using this discretization, and using $\varkappa = \rho\kappa$, the diffusion equations become

$$\frac{1}{c} \frac{\partial I_k}{\partial t} - \nabla \cdot D_k \nabla I_k + \varkappa_k I_k = \varkappa_k \beta_k(T) \quad (2a)$$

$$\rho C_v \frac{\partial T}{\partial t} = \nabla \cdot K \nabla T - \nabla \mathbb{P} u + \sum_{g=1}^{N_g} \varkappa_g [I_g - \beta_g(T)]' + Q, \quad (2b)$$

where

$$I_k = \int_{G_k} I(\nu) d\nu \quad (3a)$$

$$\beta_k = \int_{G_k} \beta(\nu, T) d\nu \quad (3b)$$

$$D_k \nabla I_k = \int_{G_k} D(\nu) \nabla I(\nu) d\nu \quad (3c)$$

$$\varkappa_k = \frac{\int_{G_k} [\beta(\nu) - I(\nu)] \varkappa(\nu) d\nu}{\int_{G_k} [\beta(\nu) - I(\nu)] d\nu}. \quad (3d)$$

Discretizing in time using Backward Euler (time index n), with lagged coefficients, Equation 1 becomes:

$$\frac{1}{c} \frac{I^{n+1} - I^n}{\Delta t^n} - \nabla \cdot D_k^n \nabla I_k^{n+1} + \varkappa_k^n I_k^{n+1} = \varkappa_k^n \beta_k(T^{n+1}); \quad k = 1 \dots N_g \quad (4a)$$

$$\rho C_v^n \frac{T^{n+1} - T^n}{\Delta t^n} = \nabla \cdot K^n \nabla T^{n+1} - \nabla \mathbb{P}u + \sum_{g=1}^{N_g} \varkappa_g^n [I_g^{n+1} - \beta_g(T^{n+1})] + Q^{n+1}. \quad (4b)$$

The time-advanced Planck function β is approximated using the first-order Taylor expansion $\beta(T^{n+1}) \approx \beta(T^n) + \frac{\partial \beta}{\partial T}(T^{n+1} - T^n)$, which leads to the system for T^{n+1} and I^{n+1} ,

$$\frac{1}{c} \frac{I^{n+1} - I^n}{\Delta t^n} - \nabla \cdot D_k^n \nabla I_k^{n+1} + \varkappa_k^n I_k^{n+1} = \varkappa_k^n \left(\beta(T^n) + \frac{\partial \beta}{\partial T} \Delta T^{n+1} \right); \quad k = 1 \dots N_g \quad (5a)$$

$$\rho C_v^n \frac{\Delta T^{n+1}}{\Delta t^n} = \nabla \cdot K^n \nabla T^{n+1} - \nabla \mathbb{P}u + \sum_{g=1}^{N_g} \varkappa_g^n \left[I_g^{n+1} - \beta(T^n) - \frac{\partial \beta}{\partial T} \Delta T^{n+1} \right] + Q^{n+1}. \quad (5b)$$

Finally, the temperature operator is split into the hydrodynamic and radiative contributions,

$$\rho C_v^n \frac{\Delta T^{n+1}}{\Delta t^n} = \nabla \cdot K^n \nabla T^{n+1} - \nabla \mathbb{P}u + Q^{n+1} \quad (6a)$$

$$\frac{1}{c} \frac{I^{n+1} - I^n}{\Delta t^n} - \nabla \cdot D_k^n \nabla I_k^{n+1} + \varkappa_k^n I_k^{n+1} = \varkappa_k^n \left(\beta(T^n) + \frac{\partial \beta}{\partial T} \Delta T^{n+1} \right); \quad k = 1 \dots N_g \quad (6b)$$

$$\rho C_v^n \frac{\Delta T^{n+1/2}}{\Delta t^n} = \sum_{g=1}^{N_g} \varkappa_g^n \left[I_g^{n+1} - \beta(T^n) - \frac{\partial \beta}{\partial T} \Delta T^{n+1/2} \right], \quad (6c)$$

where $T^{n+1} = T^n + \Delta T^{n+1/2} + \Delta T^{n+1}$. The indices $n+1/2$ and $n+1$ in this case refer to two separate contributions in the same time step, not temperature change at an intermediate time step.

Equation 6a can be immediately solved for temperature change,

$$\Delta T^{n+1/2} = \frac{\sum_{g=1}^{N_g} \kappa_g [I_g^{n+1} - \beta_g^n] + Q}{\frac{\rho C_v}{\delta t^n} + \sum_{g=1}^{N_g} \kappa_g \frac{\partial \beta}{\partial T}}, \quad (7)$$

which is then substituted into Equation 6b to give the radiative diffusion with implicit ΔT :

$$\frac{1}{c\Delta t^n} (I^{n+1} - I^n) - \nabla \cdot D_k^n \nabla I_k^{n+1} + \kappa_k^n I_k^{n+1} = \eta \chi_k \sum_{g=1}^{N_g} \kappa_g I_g^{n+1} + q_k \quad (8a)$$

where

$$\eta = \left[\sum_{g=1}^{N_g} \kappa_g \frac{\partial \beta_g}{\partial T} \right] \cdot \left[\frac{\rho C_v}{\Delta t^n} + \sum_{g=1}^{N_g} \kappa_g \frac{\partial \beta_g}{\partial T} \right]^{-1} \quad (8b)$$

$$\chi_k = \left[\kappa_k \frac{\partial \beta_k}{\partial T} \right] \cdot \left[\sum_{g=1}^{N_g} \kappa_g \frac{\partial \beta_g}{\partial T} \right]^{-1} \quad (8c)$$

$$q_k = \kappa_k \beta_k + \eta \chi_k \left[Q - \sum_{g=1}^{N_g} \kappa_g \frac{\partial \beta_g}{\partial T} \right]. \quad (8d)$$

Equation 8a is solved for intensity, then temperature change is recovered afterward. This can be rearranged to further resemble a steady-state neutron diffusion equation,

$$-\nabla \cdot D_k \nabla I_k^{n+1} + (\sigma_a + \sigma_{f,k}) I_k^{n+1} = \eta \chi_k \sum_{g=1}^{N_g} \sigma_{f,g} I_g^{n+1} + S_k \quad (9)$$

where

$$\sigma_a = \frac{1}{c\Delta t^n} \quad (10a)$$

$$\sigma_{f,k} = \kappa_k \quad (10b)$$

$$S_k = q_k + \frac{I_k^n}{c\Delta t^n}. \quad (10c)$$

Here $\sigma_{f,k} = \kappa_k$ acts as a fission cross section, where photons are re-emitted with the “fission” spectrum χ_k , with multiplicity η , and capture cross section σ_a . The physical interpretation is that photons may be lost to material absorption or streaming in time (from $\frac{d}{dt}$). Those photons absorbed in the material are either re-emitted in the same time step, or contribute to temperature gain in the material.

The intensity equation is solved using power iteration on the fission source. Starting from initial iterate $I_k^{(0)}$, the $l + 1$ iterate for intensity at the $n + 1$ time step is found by solving the linear system

$$-\nabla \cdot D_k \nabla I_k^{l+1} + (\sigma_a + \sigma_{f,k}) I_k^{l+1} = \eta \chi_k \sum_{g=1}^{N_g} \sigma_{f,g} I_g^l + S_k. \quad (11)$$

The full iterative algorithm is found in Algorithm 1.

Algorithm 1 Unaccelerated Iteration Algorithm

```

while  $t^n \leq t^{end}$  do
   $n \leftarrow n + 1$ 
  Compute (lagged) temperature-dependent coefficients using  $T|_{t=t^{n-1}}$ 
  while  $\|I^{(l)} - I^{(l-1)}\| > \epsilon \|I^{(l-1)}\|$  do
     $l \leftarrow l + 1$ 
     $I_k^{(l+1/2)} \leftarrow$  Source iteration on  $\eta \chi_k \sum_{g=1}^{N_g} \sigma_{f,g} I_g^{n+1}$  (11)
  end while
   $I_k^n \leftarrow I_k^{(l+1)}$ 
  Recover  $\Delta T^{n+1/2}$  from  $I_k^n$  (7)
  if  $K \neq 0 \wedge u \neq 0$  then
    Solve thermal diffusion equation for  $\Delta T^{n+1}$ 
  end if
   $T^{n+1} \leftarrow T^n + \Delta T^{n+1/2} + \Delta T^{n+1}$ 
end while

```

1.1 Synthetic Acceleration Method

The equation for exact multigroup error of the l th iterate (Equation 11) is

$$-\nabla \cdot D_k \nabla \epsilon_k^l + (\sigma_a + \sigma_{f,k}) \epsilon_k^l = \eta \chi_k \sum_{g=1}^{N_g} \sigma_{f,g} [\epsilon_g^l + I_g^l - I_g^{l-1}]. \quad (12)$$

At equilibrium, this becomes

$$(\sigma_a + \sigma_{f,k}) \epsilon_k^l = \eta \chi_k \sum_{g=1}^{N_G} \sigma_{f,g} \epsilon_g^l + \eta \chi_k \sum_{g=1}^{N_G} \sigma_{f,g} (I_g^l - I_g^{l-1}) \quad (13)$$

and the group spectrum of ϵ is

$$\frac{\epsilon_k^l}{\mathcal{E}^l} = \frac{\langle \sigma_t \rangle \chi_k}{\sigma_{t,k}} \quad (14)$$

where

$$\mathcal{E}^l = \sum_{g=1}^{N_g} \epsilon_g^l \quad (15a)$$

$$\langle \sigma_t \rangle = \left[\sum_{g=1}^{N_g} \frac{\chi_g}{\sigma_{t,g}} \right]^{-1}. \quad (15b)$$

Then, all groups of Equation 12 are summed, using this spectrum, to get

$$-\nabla \cdot [\langle D \rangle \nabla \mathcal{E}^l + \langle D' \rangle \mathcal{E}^l] + \mathcal{E}^l [\sigma_a + \langle \sigma_f \rangle (1 - \eta)] = \eta r^l \quad (16)$$

where

$$\langle D \rangle = \sum_{k=1}^{N_g} D_k \frac{\langle \sigma_t \rangle \chi_k}{\sigma_{t,k}} \quad (17a)$$

$$\langle D' \rangle = \sum_{k=1}^{N_g} D_k \nabla \frac{\langle \sigma_t \rangle \chi_k}{\sigma_{t,k}} \quad (17b)$$

$$\langle \sigma_f \rangle = \langle \sigma_t \rangle \sum_{g=1}^{N_g} \frac{\chi_g \sigma_{f,g}}{\sigma_{t,g}} \quad (17c)$$

$$r^{l+1/2} = \sum_{g=1}^{N_g} \sigma_{f,g} (I_g^{l+1/2} - I_g^l). \quad (17d)$$

Then the spectrum from Equation 14 can be used to calculate the groupwise correction terms from the grey correction in Equation 16. The accelerated iterative scheme becomes

$$-\nabla \cdot D_k \nabla I_k^{l+1} + (\sigma_a + \sigma_{f,k}) I_k^{l+1} = \eta \chi_k \sum_{g=1}^{N_g} \sigma_{f,g} I_g^l + S_k \quad (18a)$$

$$-\nabla \cdot [\langle D \rangle \nabla \mathcal{E}^{l+1/2} + \langle D' \rangle \mathcal{E}^{l+1/2}] + \mathcal{E}^{l+1/2} [\sigma_a + \langle \sigma_f \rangle (1 - \eta)] = \eta r^{l+1/2} \quad (18b)$$

$$I^{l+1} = I_k^{l+1} + \mathcal{E}^{l+1/2} \frac{\langle \sigma_t \rangle \chi_k}{\sigma_{t,k}} \quad (18c)$$

The accelerated iterative algorithm is displayed in Algorithm 2

Algorithm 2 Accelerated Iteration Algorithm

```

while  $t^n \leq t^{end}$  do
   $n \leftarrow n + 1$ 
  Compute (lagged) temperature-dependent coefficients using  $T|_{t=t^{n-1}}$ 
  while  $\|I^{(l)} - I^{(l-1)}\| > \epsilon \|I^{(l-1)}\|$  do
     $l \leftarrow l + 1$ 
     $I_k^{(l+1/2)} \leftarrow$  Source iteration on  $\eta\chi_k \sum_{g=1}^{N_g} \sigma_{f,g} I_g^{n+1}$ , using (18a)
     $r^{(l+1/2)} \leftarrow \sum_{g=1}^{N_g} \sigma_{f,g} \left( I_g^{l+1/2} - I_g^l \right)$ 
     $\mathcal{E}^l \leftarrow$  Grey correction equation using (18b)
     $I_k^{(l)} \leftarrow I_k^{(l+1/2)} + \mathcal{E}^{l+1/2} \frac{\langle \sigma_t \rangle \chi_k}{\sigma_{t,k}}$  using (18c)
  end while
   $I_k^n \leftarrow I_k^{(l+1)}$ 
  Recover  $\Delta T^{n+1/2}$  from  $I_k^n$ 
  if  $K \neq 0 \wedge u \neq 0$  then
    Solve thermal diffusion equation for  $\Delta T^{n+1}$ 
  end if
   $T^{n+1} \leftarrow T^n + \Delta T^{n+1/2} + \Delta T^{n+1}$ 
end while

```

2. Numerical Implementation

Using the multigroup diffusion equation with Backward Euler, lagged coefficients, and implicit temperature to approximate a 1D thermal radiative transfer (TRT) problem gives, at each time step, Equation 11 can be split into two first-order ODEs,

$$\frac{\partial F_k^{n+1}}{\partial x} + (\sigma_a + \sigma_{f,k}) I_k^{l+1} = \eta\chi_k \sum_{g=1}^{N_g} \sigma_{f,g} I_g^l + S_k \quad (19a)$$

$$F_k^{n+1} = -D_k \frac{\partial I_k^{n+1}}{\partial x} \quad (19b)$$

$$0 \leq x \leq X, \quad k = 1, \dots, N_g$$

where temperature is recovered using Equation 7. Boundary conditions for these equations are given by

$$I^+(0) = I_{in}^+ \quad (20)$$

$$I^-(X) = I_{in}^- \quad (21)$$

$$F^+(0) = F_{in}^+ \quad (22)$$

$$F^-(X) = F_{in}^- \quad (23)$$

where

$$I^\pm = \pm \int_0^{\pm 1} \mathcal{I}(\mu) d\mu \quad (24)$$

$$F^\pm = \pm \int_0^{\pm 1} \mu \mathcal{I}(\mu) d\mu \quad (25)$$

First, the spatial domain is divided into N_x cells $K_i = [x_{i-1/2}, x_{i+1/2}]$; $i = 1, \dots, N_x$, where $x_{1/2} = 0$ and $x_{N_x+1/2} = X$, and in each cell K_i the width is $\Delta x_i = x_{i+1/2} - x_{i-1/2}$. Then, if temperature (and temperature-dependent coefficients) are taken to be constant within each cell, then, in the accelerated correction equation, the term $\langle D' \rangle$ becomes zero, and the grey correction equation has the same form as Equation 19, so they may be solved in the same way.

2.1 LD Formulation

First, I and F (and the source S_k) from Equation 19 are assumed to take the form

$$I(x) = I_{L,i} b_{L,i} + I_{R,i} b_{R,i} \quad (26a)$$

$$F(x) = F_{L,i} b_{L,i} + F_{R,i} b_{R,i} \quad (26b)$$

$$S(x) = S_{L,i} b_{L,i} + S_{R,i} b_{R,i} \quad (26c)$$

$$b_{L,i} = \frac{x_{i+1/2} - x}{\Delta x_i}, \quad b_{R,i} = \frac{x - x_{i-1/2}}{\Delta x_i} \quad (26d)$$

where the multigroup index k and the time index n have been omitted without loss of generality. The weak form of Equation 19 is found by multiplying by the test functions $b_{L,i}$ and $b_{R,i}$ and integrating over the cell.

$$\begin{aligned} \int_{x_{i-1/2}}^{x_{i+1/2}} \frac{dF^{l+1}}{dx} b_{c,i} dx + (\sigma_a + \sigma_f) \int_{x_{i-1/2}}^{x_{i+1/2}} I^{l+1} b_{c,i} dx = \\ \eta \chi \sum_{g=1}^{N_g} \sigma_{f,g} \int_{x_{i-1/2}}^{x_{i+1/2}} I_g^{(l)} b_{c,i} dx + \int_{x_{i-1/2}}^{x_{i+1/2}} S b_{c,i} dx \end{aligned} \quad (27a)$$

$$\begin{aligned} \int_{x_{i-1/2}}^{x_{i+1/2}} F^{l+1} b_{c,i} dx = -D \int_{x_{i-1/2}}^{x_{i+1/2}} \frac{dI^{l+1}}{dx} b_{c,i} dx \\ c = L, R; \quad i = 1, \dots, N_x \end{aligned} \quad (27b)$$

or, rewriting the $\frac{d}{dx}$ terms using integration by parts,

$$[F^{l+1}b_{c,i}]_{x_{i-1/2}}^{x_{i+1/2}} - \int_{x_{i-1/2}}^{x_{i+1/2}} \frac{db_{c,i}}{dx} F^{l+1} dx + (\sigma_a + \sigma_f) \int_{x_{i-1/2}}^{x_{i+1/2}} I^{l+1} b_{c,i} dx = \eta \chi \sum_{g=1}^{N_g} \sigma_{f,g} \int_{x_{i-1/2}}^{x_{i+1/2}} I_g^{(l)} b_{c,i} dx + \int_{x_{i-1/2}}^{x_{i+1/2}} S b_{c,i} dx \quad (28a)$$

$$\int_{x_{i-1/2}}^{x_{i+1/2}} F^{l+1} b_{c,i} dx = -D \left([I^{l+1} b_{c,i}]_{x_{i-1/2}}^{x_{i+1/2}} - \int_{x_{i-1/2}}^{x_{i+1/2}} \frac{db_{c,i}}{dx} I^{l+1} dx \right) \quad (28b)$$

$c = L, R; \quad i = 1, \dots, N_x.$

Using the assumed forms in Equation 26 and the interface conditions

$$I^b(x_{i+1/2}) = I^+ + I^- = \frac{1}{2} (I_{R,i} + I_{L,i+1}) + \frac{3}{4} (F_{R,i} - F_{L,i+1}) \quad (29)$$

$$F^b(x_{i+1/2}) = F^+ + F^- = \frac{1}{4} (I_{R,i} - I_{L,i+1}) + \frac{1}{2} (F_{R,i} + F_{L,i+1}) \quad (30)$$

the local equations on the interior become,

$$\frac{1}{4} (I_{L,i}^{(l+1)} - I_{R,i-1}^{(l+1)}) + \frac{1}{2} (F_{R,i}^{(l+1)} - F_{R,i-1}^{(l+1)}) + \frac{\sigma_t \Delta x_i}{6} (m_{LL} I_{L,i}^{(l+1)} + m_{LR} I_{R,i}^{(l+1)}) = \eta \chi \sum_{g=1}^{N_g} \frac{\sigma_{f,g} \Delta x_i}{6} (m_{LL} I_{g,L,i}^{(l)} + m_{LR} I_{g,R,i}^{(l)}) + \frac{\Delta x_i}{6} (m_{LL} S_{L,i} + m_{LR} S_{R,i}) \quad (31)$$

$$\frac{1}{4} (I_{R,i}^{(l+1)} - I_{L,i+1}^{(l+1)}) + \frac{1}{2} (F_{L,i+1}^{(l+1)} - F_{L,i}^{(l+1)}) + \frac{\sigma_t \Delta x_i}{6} (m_{RL} I_{g,L,i}^{(l+1)} + m_{RR} I_{g,R,i}^{(l+1)}) = \eta \chi \sum_{g=1}^{N_g} \frac{\sigma_{f,g} \Delta x_i}{6} (m_{RL} I_{L,i}^{(l)} + m_{RR} I_{R,i}^{(l)}) + \frac{\Delta x_i}{6} (m_{RL} S_{L,i} + m_{RR} S_{R,i}) \quad (32)$$

$$\frac{\Delta x_i}{6} (m_{LL} F_{L,i} + m_{LR} F_{R,i}) + D \left[\frac{3}{4} (F_{L,i} - F_{R,i-1}) + \frac{1}{2} (I_{R,i} - I_{R,i-1}) \right] = 0 \quad (33)$$

$$\frac{\Delta x_i}{6} (m_{RL} F_{L,i} + m_{RR} F_{R,i}) + D \left[\frac{3}{4} (F_{R,i} - F_{L,i+1}) + \frac{1}{2} (I_{L,i+1} - I_{L,i}) \right] = 0 \quad (34)$$

and on the boundaries,

$$\frac{1}{4} (I_{L,i}^{(l+1)}) + \frac{1}{2} (F_{R,i}^{(l+1)}) + \frac{\sigma_t \Delta x_i}{6} (m_{LL} I_{L,i}^{(l+1)} + m_{LR} I_{R,i}^{(l+1)}) = \eta \chi \sum_{g=1}^{N_g} \frac{\sigma_{f,g} \Delta x_i}{6} (m_{LL} I_{g,L,i}^{(l)} + m_{LR} I_{g,R,i}^{(l)}) + \frac{\Delta x_i}{6} (m_{LL} S_{L,i} + m_{LR} S_{R,i}) + F^{in,+} \quad (35)$$

$$\frac{1}{4} \left(I_{R,i}^{(l+1)} \right) - \frac{1}{2} \left(F_{L,i}^{(l+1)} \right) + \frac{\sigma_t \Delta x_i}{6} \left(m_{RL} I_{L,i}^{(l+1)} + m_{RR} I_{R,i}^{(l+1)} \right) =$$

$$\eta \chi \sum_{g=1}^{N_g} \frac{\sigma_{f,g} \Delta x_i}{6} \left(m_{RL} I_{g,L,i}^{(l)} + m_{RR} I_{g,R,i}^{(l)} \right) + \frac{\Delta x_i}{6} (m_{RL} S_{L,i} + m_{RR} S_{R,i}) - F^{in,-} \quad (36)$$

$$\frac{\Delta x_i}{6} (m_{LL} F_{L,i} + m_{LR} F_{R,i}) + D \left[\frac{3}{4} (F_{L,i}) + \frac{1}{2} (I_{R,i}) \right] = DI^{in,+} \quad (37)$$

$$\frac{\Delta x_i}{6} (m_{RL} F_{L,i} + m_{RR} F_{R,i}) + D \left[\frac{3}{4} (F_{R,i}) - \frac{1}{2} (I_{L,i}) \right] = -DI^{in,-}. \quad (38)$$

Written in matrix form, these equations become, on the interior,

$$\begin{bmatrix} \mathbf{B}_1 + \sigma \Delta x \overline{\mathbf{M}} & \mathbf{B}_2 \\ D\mathbf{B}_2 & \Delta x \overline{\mathbf{M}} + 3D\mathbf{B}_1 \end{bmatrix} \begin{bmatrix} \vec{I} \\ \vec{F} \end{bmatrix} = \begin{bmatrix} \mathbf{M}\vec{Q} \\ 0 \end{bmatrix} \quad (39a)$$

where

$$\mathbf{B}_1 = \begin{bmatrix} -1/4 & 1/4 & 0 & 0 \\ 0 & 0 & 1/4 & -1/4 \end{bmatrix} \quad (39b)$$

$$\mathbf{B}_2 = \begin{bmatrix} -1/2 & 0 & 1/2 & 0 \\ 0 & -1/2 & 0 & 1/2 \end{bmatrix} \quad (39c)$$

$$\mathbf{M} = \frac{1}{6} \begin{bmatrix} m_{LL} & m_{LR} \\ m_{RL} & m_{RR} \end{bmatrix} \quad (39d)$$

$$\overline{\mathbf{M}} = \frac{1}{6} \begin{bmatrix} 0 & m_{LL} & m_{LR} & 0 \\ 0 & m_{RL} & m_{RR} & 0 \end{bmatrix} \quad (39e)$$

$$\vec{I} = \begin{pmatrix} I_{R,i-1} \\ I_{L,i} \\ I_{R,i} \\ I_{L,i+1} \end{pmatrix}, \quad \vec{F} = \begin{pmatrix} F_{R,i-1} \\ F_{L,i} \\ F_{R,i} \\ F_{L,i+1} \end{pmatrix}, \quad \vec{Q} = \begin{pmatrix} \eta \chi \sum_g \sigma_{f,g} I_{g,L,i} + S_{L,i} \\ \eta \chi \sum_g \sigma_{f,g} I_{g,R,i} + S_{R,i} \end{pmatrix} \quad (39f)$$

On the left and right boundaries, the left- or right-most column of each local matrix is truncated, with the boundary conditions added to the source:

$$\begin{bmatrix} \mathbf{B}_1^c + \sigma \Delta x \overline{\mathbf{M}}^c & \mathbf{B}_2^c \\ D\mathbf{B}_2^c & \Delta x \overline{\mathbf{M}}^c + 3D\mathbf{B}_1^c \end{bmatrix} \begin{bmatrix} \vec{I}^c \\ \vec{F}^c \end{bmatrix} = \begin{bmatrix} \mathbf{M}\vec{Q} \\ 0 \end{bmatrix} + \begin{bmatrix} \vec{b}_F \\ D\vec{b}_I \end{bmatrix}, \quad c = L, R \quad (40)$$

where, for example,

$$\mathbf{B}_1 = \begin{bmatrix} \cancel{-1/4} & 1/4 & 0 & 0 \\ \emptyset & 0 & 1/4 & -1/4 \end{bmatrix} = \mathbf{B}_1 = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/4 & -1/4 \end{bmatrix}, \text{ etc,} \quad (41)$$

and the boundary vectors are

$$\vec{b}_F = \begin{bmatrix} \delta_{c,L} F^{in,+} \\ -\delta_{c,R} F^{in,-} \end{bmatrix}, \quad \vec{b}_I = \begin{pmatrix} \delta_{c,L} I^{in,+} \\ -\delta_{c,R} I^{in,-} \end{pmatrix}. \quad (42)$$

The global matrix is then a $(4N_x \times 4N_x)$ square matrix. In the Python implementation, the first $2N_x$ rows of the global matrix are composed of the first two rows of the local matrix (the zeroth moment equation), and the last $2N_x$ rows of the global matrix are composed of the last two rows of the local matrix (Fick's Law), so that the global system is of the form

$$\mathbf{G} \begin{bmatrix} I \\ F \end{bmatrix} = \begin{bmatrix} Q + b_F \\ b_I \end{bmatrix}. \quad (43)$$

The global matrix assignment is performed by `method.assemble_global_matrix()`, and the matrix definition can be found in `tools`. For example, the interior assembly:

```
# interior elements
for
    1      1
    2      1
    2      2
    2

# zeroth moment, intensity
    2      2      2
                                1
# zeroth moment, flux
    2      2      2
                                1

# first moment, intensity
    2              2      2
                                1
# first moment, flux
    2              2      2
                                1
                                3
```

2.2 Unaccelerated Iteration

2.3 Accelerated Iteration

2.4 Fourier Analysis and Convergence

3. Tests

3.1 Fleck and Cummings Test

3.2 Test A1, nonzero boundary conditions

3.3 Test A2, one group

3.4 Fourier-informed tests

References

- [1] J. E. Morel, E. W. Larsen, and M. Matzen, “A synthetic acceleration scheme for radiative diffusion calculations,” *Journal of Quantitative Spectroscopy and Radiative Transfer*, vol. 34, no. 3, pp. 243–261, 1985.